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# BTDU — Blackmore Technology Data Universe

Authoritative knowledge, data atoms, semantic relationships and exact reconstruction

BTG-LIB-BTDU-SG · Edition 1.0 · Audience: Knowledge custodians, researchers, records managers, rights holders and institutional reviewers · Issued 23 October 2026

**Publication boundary:** This is an explanation of the system and its public operating model, not a source release, engineering specification, certification or promise of live service availability. Product status is stated with the limitations below.

## Executive overview

BTDU is BTG's governed reconstructive knowledge substrate. Its guiding principle is "One authoritative payload. Unlimited semantic views." Rather than treating each interpretation as a separate uncontrolled copy, it distinguishes stored authoritative material from indexes, relationships and views that describe how the material may be used.

## 01 — What the Data Universe means

BTDU treats information as a structured universe of identifiable components and relationships. A reader can think of data atoms as the smallest identified parts needed by a representation, with larger structures expressing relationships, versions or views.

The model is intended to preserve exact authoritative material and its lineage while letting authorized users ask different questions about it. A semantic view is not a right to rewrite the original evidence.

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## 02 — One authoritative payload

Keeping one authoritative representation aims to reduce contradiction when several applications require different interpretations of the same information. The same retained record may support a research view, a rights view or a market disclosure view, subject to permission.

Where a document or dataset changes, the new version must be distinguishable from the prior one. A view that omits detail for privacy is not proof that those underlying details do not exist.

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## 03 — Atoms, relationships and reconstruction

The public design describes exact byte components, shared references, formula-like structures, indexes and lineage relationships. Reconstruction should be evaluated against original-byte checksums for a bounded corpus and documented environment.

Not every possible data type will produce the same compression result. A benchmark using particular inputs and a controlled implementation must not be extrapolated to universal storage savings.

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## 04 — Semantic views and purpose limits

Research users may need mathematical relationships; an owner may need origin and permissions; a reviewer may need a chronology. BTDU is intended to produce those differing views without declaring each a new authoritative asset.

Access decisions occur outside mere lookup convenience. ENTITY-authorized rights and relevant purpose policies govern who can see protected records and what may be exported.

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## 05 — Preservation and recovery

A defensible preservation process identifies the source, format, exact-byte hash, version, dependencies and a means to reconstruct a usable artifact. Repeated verification should test that the result matches the original rather than merely looks similar.

Cold recovery should assume that some devices, indexes or network services may be unavailable. Whether recovery is successful depends on the exact backups, remaining keys, storage media and tested restoration procedure.

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## 06 — Lineage and attribution

Lineage records why a dataset, finding or component exists and from which authorized inputs it arose. Attribution may recognize a source or contribution, while legal rights require separate agreements and competent authority.

A cryptographic manifest records integrity and referenced assertions. It does not automatically prove an external identity or resolve a disputed copyright claim.

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## 07 — Reuse by NIKI and Circuits

NIKI may consult an authorized BTDU knowledge view while proposing research. A DataChip or Circuit can reference the exact version of relevant inputs and outputs, allowing future tests to understand which material was used.

Retaining qualified solutions should not erase failed experiments or permit a changed model to claim old results. Claims about novel invention require independent qualification.

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## 08 — Relationship to ADAM and ENTITY

ENTITY governs identity, rights and evidence. ADAM applies authorized state-transition policies. BTDU maintains governed knowledge representations and reconstruction. Each system has a distinct role and should not silently assume the authority of another.

An operator of a storage provider should not acquire the ability to grant IP ownership or issue a financial instrument merely by hosting a payload.

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## 09 — Public disclosure and private material

A public market description may disclose dataset purpose, tested utility, scope, version and licensing terms while withholding personal rows, source code, protected research or third-party licensed content.

The Exchange should only link to the releasable information view. Bridge consent does not itself authorize full reproduction of source records to prospective buyers.

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## 10 — Evidence and performance boundaries

BTDU publicly reports controlled storage and exact-recovery results under named versions and datasets. Readers must distinguish bounded tests from universal proof of compression, security, availability or independent replication.

Before production adoption, reviewers need release-specific evidence for recovery, integrity, permissions, scaling and operational support. This document deliberately does not reveal proprietary reconstruction formulas.

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## Related manuals and use

The companion User & Operator Manual describes the decisions and reviews applicable to this system. References to proposed user flows are explicitly conditional on qualification of the relevant product release.